Paraphrase-Consistent Extractive Question Answering on SQuAD v1.1

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Abstract

Extractive question answering (QA) models fine-tuned on SQuAD often change their answer when a question is rephrased without changing its meaning. We train extractors on each original question and a filtered EN→DE→EN back-translation of it, with a symmetric KL term between the start/end distributions of the two views, and separate the effect of the paraphrased training data from that of the KL term. On BERT-base, DeBERTa-v3-base and DeBERTa-v3-large the full recipe raises the share of paraphrase pairs answered identically by 4.6, 3.8 and 3.3 points over plain training, while SQuAD v1.1 EM and F1 stay within 0.3 points in our single-seed runs; only on BERT-base do we isolate the KL term (+1.4 of the 4.6 points, over augmentation alone). The proposed system, a DeBERTa-v3-large extractor trained with augmentation and KL, reaches 90.0 EM / 95.4 F1 with 92.7% agreement. On the human paraphrases of Gan and Ng (2019) its paraphrase F1 stays at 93.8 while agreement rises from 92.3% to 95.0%. A learned second-stage span reranker does not improve in-distribution accuracy beyond noise and is not part of the system.

1 Introduction

Fine-tuned encoders for extractive QA (Rajpurkar et al., 2016) such as BERT (Devlin et al., 2019) and DeBERTa (He et al., 2021, 2023) score highly on SQuAD, but they are *over-sensitive* to the surface form of the question, with sizeable drops on human paraphrases (Gan and Ng, 2019) and on distractor sentences (Jia and Liang, 2017).

Gan and Ng (2019) train on paraphrased questions; Xie et al. (2020) regularise the output distribution towards invariance under label-preserving perturbations (e.g. back-translation (Sennrich et al., 2016)). Consistency training itself is not new. Our question is narrower: *does explicitly regularising the start/end span distributions of an original and a paraphrased question add stability beyond simply training on the paraphrases*, and does it cost accuracy?

Our contributions are: (i) a symmetric-KL consistency objective on span distributions with filtered back-translated paraphrases, and a controlled comparison of plain training, augmentation only and augmentation plus KL, where the KL term is isolated on BERT-base only (§2, §3); (ii) its evaluation on three backbones of different strength against human paraphrases (Gan and Ng, 2019) and distractor sentences (Jia and Liang, 2017), with every decision taken on held-out training articles; (iii) negative results: optimal-transport variants (Peyré and Cuturi, 2019; Kuhn, 1955) that do not beat KL, and a second-stage span reranker in the style of RankQA (Kratzwald et al., 2019) and a training-free paraphrase ensemble that do not improve in-distribution accuracy (§3.1).

2 Method

Extractor and candidates. Each (question, passage) pair is encoded as `[CLS]` q `[SEP]` $c^{(w)}$ `[SEP]` for sliding windows $c^{(w)}$ of at most L tokens with step S; a linear head gives start and end logits, windows without the answer are labelled with the `[CLS]` position, and $\mathcal{L}_{\text{CE}}$ is the mean of the start and end cross-entropies (Devlin et al., 2019). At inference every span (i, j) with $i \le j < i + 30$ inside the passage is scored by $z_s[i] + z_e[j]$ minus the window's null score $z_s[0] + z_e[0]$; the $k = 20$ best spans with distinct normalised text are kept, and the top one is the extractor's answer.

Paraphrases. Every training question is back-translated through German with `opus-mt-en-de` and `opus-mt-de-en` (Tiedemann and Thottingal, 2020). A pair is kept when the paraphrase differs from the question after normalisation and the cosine similarity of their `all-MiniLM-L6-v2` embeddings (Reimers and Gurevych, 2019) is at least

0.85. This accepts a paraphrase for 68,439 of the 83,484 training questions (82.0%) and 8,622 of the 10,570 dev questions (81.6%); dev paraphrases are used only to measure agreement.

Consistency objective. For a question q, its paraphrase q' and a shared window, let P_t, P'_t ($t \in \{s, e\}$) be the start and end distributions restricted to the m passage tokens. We minimise

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(q) + \mathcal{L}_{\mathrm{CE}}(q') + \lambda\, \mathcal{D}_{\mathrm{KL}}^{\mathrm{sym}},$$
$$\mathcal{D}_{\mathrm{KL}}^{\mathrm{sym}} = \tfrac{1}{4} \sum_{t \in \{s,e\}} \big[\mathrm{KL}(P_t \| P'_t) + \mathrm{KL}(P'_t \| P_t)\big]. \tag{1}$$

$\lambda = 0$ is the augmentation-only baseline of Gan and Ng (2019); dropping $\mathcal{L}_{\mathrm{CE}}(q')$ as well gives the plain extractor (CE). KL ignores token order, so as negative ablations we replace it by $\lambda_{\mathrm{OT}} = 0.5$ times the 1-D Wasserstein distance between P_t and P'_t (*OT*) and add $\mu = 0.1$ times a Hungarian matching of the two views' top-5 spans with cost $0.5\, d_{\mathrm{pos}} + 0.5\,(1 - \mathrm{IoU})$, where d_{pos} is the boundary-position distance normalised by twice the window length (*dual*); these weights were not tuned.

Second-stage span selection (secondary experiment). The top-20 list contains an exactly correct span for 98.9% of the held-out early-stopping questions, so we also tried a *learned span reranker*. For each candidate it receives the extractor's span hidden states (Lee et al., 2017) and 14 scalar features (scores, rank, length and, when a paraphrase is available, three *pair features* from the paraphrase view's score for the same span). A small MLP scores each candidate and is trained with listwise softmax cross-entropy (Cao et al., 2007) against graded targets on the held-out training articles only (758 for early stopping). We test it with and without the pair features, with its arg-max replacing the extractor's answer or added to the extractor's log-probability (*blend*, weight α), and trained with a symmetric-KL term between its distributions for q and q' (*consistency-trained*, weight w). The *training-free paraphrase ensemble* takes the candidate maximising $\log p(\cdot \mid q) + \beta \log p(\cdot \mid q')$ over the extractor's own scores. Both the pair-feature reranker and the ensemble need a paraphrase at inference time.

Protocol. We hold out 5% of the training *articles* (23 of 442; 4,115 questions) for every decision: the best checkpoint of a run is chosen on a fixed 2,000-question sample of them, λ on their F1 and agreement, and the reranker is trained on them. The dev and external sets are used only for final numbers, with the official `evaluate-v2.0.py`. *Agreement* is the percentage of (question, paraphrase) pairs whose normalised answers are identical; since always returning the same wrong span would also agree, we also report *both correct*, the share of pairs where both answers are exact matches (available for the proposed extractor, whose paraphrase-view predictions were saved), and read agreement together with EM/F1 and with F1 on human paraphrases, which is scored against gold answers. The λ rule was fixed before training: the largest $\lambda \in \{0.5, 1, 2\}$ whose held-out F1 is within 0.5 points of the best. Second-stage differences, and the DeBERTa-v3-large CE-vs-KL differences on dev EM/F1, carry 95% paired percentile-bootstrap intervals over questions (pairs for agreement); they quantify evaluation-sample uncertainty only, not training-seed variability.

3 Experiments

Setup. Backbones: `bert-base-uncased`, DeBERTa-v3-base and DeBERTa-v3-large, all with $L = 384$, $S = 128$, AdamW, 10% linear warm-up, 2 epochs and the best held-out-F1 checkpoint; learning rate 3×10^{-5}, batch 16 (base models) or 10^{-5}, batch 8 with two-step accumulation (large), bf16. Each run was trained once on one H100. **Test sets**: SQuAD v1.1 dev (10,570 questions) with back-translated paraphrases for agreement; the paraphrased dev questions of Gan and Ng (2019) (1,062 non-adversarial and 56 adversarial, each paired with its original question); and AddSent (3,560) and AddOneSent (1,787) of Jia and Liang (2017), again with back-translated paraphrases.

Main results. Table 1 gives the extractor-alone results. *Agreement rises on every backbone.* Relative to plain CE, the full recipe (paraphrase augmentation + KL, $\lambda = 1$) gives +4.6 agreement on BERT-base (83.3% $\rightarrow$ 87.9%), +3.8 on DeBERTa-v3-base (88.0% $\rightarrow$ 91.8%) and +3.3 on DeBERTa-v3-large (89.4% $\rightarrow$ 92.7%). *Augmentation and KL are separated only on BERT-base:* augmentation alone gives 86.5% (+3.2) and the KL term a further +1.4; for the two DeBERTa models there is no augmentation-only run, so their gains cannot be split. *Accuracy changes little:* the seven BERT-base runs lie within 80.9–81.4 EM and 88.2–88.6 F1; DeBERTa-v3-base moves by -0.2 F1 and DeBERTa-v3-large by

		SQuAD dev			AddSent		AddOneSent	G&N paraphrases			G&N adv.
Backbone	**Objective**	EM	F1	Agr.	F1	Agr.	F1	$F1_{orig}$	$F1_{para}$	Agr.	F1
BERT-base	CE	81.1	88.5	83.3	64.9	77.9	73.9	88.4	85.6	89.1	58.6
BERT-base	CE + aug.	81.0	88.2	86.5	65.9	82.1	73.8	87.4	85.7	91.4	54.1
BERT-base	CE + aug. + KL	81.4	88.6	87.9	66.1	84.1	74.5	87.5	86.1	91.0	63.6
DeBERTa-v3-base	CE	88.3	93.9	88.0	85.1	85.5	88.7	93.5	91.3	91.8	81.9
DeBERTa-v3-base	CE + aug. + KL	88.3	93.7	91.8	84.2	90.2	87.9	92.4	91.0	93.7	82.9
DeBERTa-v3-large	CE	90.3	95.5	89.4	86.0	84.1	89.3	94.9	93.8	92.3	85.6
DeBERTa-v3-large	**CE + aug. + KL**	**90.0**	**95.4**	**92.7**	**87.0**	**90.4**	**90.2**	**94.9**	**93.8**	**95.0**	**90.3**

Table 1: Extractor alone, official scorer. *Agr.*: share of (question, paraphrase) pairs with the same normalised answer; back-translated paraphrases on dev (8,622 pairs) and AddSent (3,007), human paraphrase vs. original question on the Gan and Ng (2019) sets (1,061 pairs; 56 adversarial questions). Bold: proposed system.

-0.3 EM / -0.1 F1. For DeBERTa-v3-large, the paired bootstrap over the 10,570 dev questions gives ΔEM -0.28 ($[-0.69, +0.14]$) and ΔF1 -0.12 ($[-0.36, +0.13]$), consistent with no change; these intervals ignore training-seed variance. Δagreement could not be bootstrapped (the CE runs' paraphrase-view predictions were not saved). For the proposed system both answers are exact for 86.6% of dev pairs (no CE counterpart). Backbone choice dominates the accuracy differences in our experiments (88.5, 93.9 and 95.5 F1 for the plain extractors). Plain BERT-base (81.1 / 88.5) is close to the 80.8 / 88.5 of Devlin et al. (2019); the original DeBERTa-large reports 90.1 / 95.5 (He et al., 2021), a different model from our v3-large, so this is context only.

Human paraphrases. The Gan and Ng (2019) sets are the most direct test of our question, since the paraphrases are written by people and not generated by our pipeline. DeBERTa-v3-large scores 94.9 F1 on the original questions and 93.8 on their paraphrases both with CE and with augmentation + KL, but agreement rises from 92.3% to 95.0% (1,061 pairs): the model answers the two phrasings more alike without losing paraphrase F1 in this run. On BERT-base, rephrasing costs CE 2.8 F1 and the full recipe 1.4 (86.1 vs. 87.5), with agreement 89.1% $\rightarrow$ 91.0%.

Selecting λ; ablations. On the 2,000 held-out selection questions, $\lambda = 0.5$, 1 and 2 give F1 82.5, 82.1 and 81.7 with agreement 89.5%, 90.6% and 91.0%: agreement rises with λ while F1 falls by a few tenths. The rule selects $\lambda = 1$, reused for DeBERTa. With untuned weights, OT and dual reach 86.3% and 86.7% agreement against 86.5% for augmentation alone and 87.9% with KL, so they are negative ablations and we keep KL.

Distractor robustness. The plain BERT-base extractor falls from 88.5 F1 on dev to 64.9 on AddSent and 73.9 on AddOneSent. All six paraphrase-trained BERT-base runs lie within 65.7–66.1 F1 on AddSent, and augmentation alone gives +1.0 of the +1.2, so we attribute the F1 change mostly to the paraphrase data; what differs is agreement (AddSent 77.9% $\rightarrow$ 84.1%). DeBERTa-v3-base (85.1 F1) loses 0.9 F1 under the full recipe, whereas DeBERTa-v3-large gains 1.0 on AddSent (86.0 $\rightarrow$ 87.0, agreement 84.1% $\rightarrow$ 90.4%) and 0.9 on AddOneSent (89.3 $\rightarrow$ 90.2). These single-run differences have opposite signs across backbones, so we report them as observations only.

3.1 Second-stage span selection

Table 2 evaluates the second stage on the proposed extractor, whose top-20 list contains an exact match for 99.5% of dev questions. The learned rerankers recover almost none of that oracle headroom: their dev EM changes of +0.2 to +0.3 all have intervals including zero and F1 moves by at most 0.1. The pair features mainly buy consistency (+1.4 agreement, $[+1.07, +1.84]$, against +0.4 without them); an earlier run gave +2.4, so one-point differences between rows are noise. Blending changes EM by tenths of a point, nothing beyond noise. Only on the distractor sets do reranker gains have intervals excluding zero (no-pair: +0.9 EM on AddSent, +1.0 on AddOneSent; +0.5 and +0.7 F1), which is out-of-distribution, not the accuracy we targeted. The training-free ensemble is the stronger consistency device: +6.6 agreement ($[+6.10, +7.16]$) and both-correct 86.6% $\rightarrow$ 89.3%, against 87.4% for the pair-feature reranker, at -0.4 dev EM ($[-0.63, -0.20]$). A reranker trained with its own consistency term does not become consistent without a paraphrase at test time. We therefore include no second stage in the proposed system; both add-ons need a paraphrase at test time.

	SQuAD dev				AddSent	AddOneSent
Second stage on the proposed extractor	ΔEM	ΔF1	ΔAgreement	Both correct	ΔF1	ΔF1
Extractor alone	–	–	–	86.6	–	–
Reranker (pair features)†	+0.2	+0.0	+1.4*	87.4	+0.6	+0.6
Reranker (no pair features)	+0.3	+0.1	+0.4*	87.0	+0.5	+0.7
Reranker (no pair), blended α=1	+0.2	+0.0	+0.3*	86.8	+0.3	+0.2
Reranker (no pair), cons.-trained w=1	+0.2	+0.0	+0.3	86.9	+0.4	+0.4
Paraphrase ensemble, training-free†	-0.4*	-0.3	+6.6*	89.3	+0.0	+0.1

Table 2: Second-stage selection among the top-20 spans of the proposed extractor. Δ = second stage − extractor; *: 95% paired-bootstrap interval excludes zero (dev EM/agreement only; no F1 intervals). *Both correct*: pairs where both answers are exact matches. †needs a paraphrase at test time.

Q: Who has the record of being the oldest quarter back in *any* Super Bowl game? **Q′:** . . . in *a* Super Bowl game?
CE: John Elway / Peyton Manning KL: Peyton Manning / Peyton Manning gold: Peyton Manning

Q: Who was the *male* singer who performed as a special guest during Super Bowl 50? **Q′:** Who was the singer who was a very special guest during Super Bowl 50?
KL: Bruno Mars / Beyoncé gold: Bruno Mars

Table 3: Dev questions, their back-translated paraphrases and the DeBERTa-v3-large answers to question / paraphrase, plain (CE) and consistency-trained (KL). Row 2: back-translation dropped "male".

4 Error Analysis

Qualitative. In the first row of Table 3 a one-word rephrasing flips the plain extractor to a plausible but wrong entity while the consistency-trained model answers both views correctly. The second row shows a remaining failure mode: back-translation dropped "male", so the paraphrase asks a different question although it passes the similarity filter. Of the 200 saved dev disagreements, 119 have exactly one view correct; we did not label their causes. Since the 0.85 filter is imperfect, 100% agreement is neither realistic nor necessarily desirable.

5 Conclusion and Limitations

A. Training extractive QA models on original/paraphrased pairs with a symmetric-KL consistency term makes them markedly more stable under paraphrasing: agreement rises by 4.6, 3.8 and 3.3 points over plain training on BERT-base, DeBERTa-v3-base and DeBERTa-v3-large, and on human paraphrases the large model's agreement rises from 92.3% to 95.0% at unchanged paraphrase F1. **B.** Ordinary SQuAD EM/F1 change negligibly (within 0.3 points) in our single-seed runs. **C.** The pattern holds across backbone strengths, but only on BERT-base can we attribute part of it to the KL term itself (+1.4 over augmentation alone, +3.2 from the paraphrase data); the DeBERTa gains are those of the full recipe. **D.** A learned span reranker gives no convincing in-distribution gain despite a 99.5% oracle, and the training-free ensemble improves consistency only with a paraphrase at test time. The proposed system is the DeBERTa-v3-large extractor with augmentation + KL, no second stage.

Limitations. Every configuration was trained once, so dev differences under half a point, the +1.4 agreement of KL over augmentation, the distractor-set differences and the 56-question adversarial set are unconfirmed; our intervals cover evaluation samples only, and none exists for agreement. Augmentation-only runs are missing for the DeBERTa models. Agreement is measured only on questions with an accepted paraphrase, and the filter admits meaning-changing paraphrases. The λ rule's 0.5-point tolerance is load-bearing (below about 0.44 it would select $\lambda = 0.5$, from 0.82 $\lambda = 2$) and was fixed before the runs but not formally pre-registered. The large model uses a different batch size and learning rate; results are for English SQuAD v1.1 only.

Contributions and external code

Written by the group: paraphrase generation and filtering, sliding-window features, the consistency objective and its OT/Hungarian variants, decoding, the span reranker and paraphrase ensemble, the hold-out protocol, and all evaluation, bootstrap and plotting code (`g17qa/`, `*.py`, `scripts/`). **External**: PyTorch, NumPy, Hugging Face Transformers with the pre-trained BERT, DeBERTa-v3 and MarianMT weights, sentence-transformers (MiniLM), scikit-learn, SciPy (Hungarian assignment), matplotlib, the official scorer `evaluate-v2.0.py` (ported in `g17qa/metrics.py`), and the test sets of Gan and Ng (2019) and Jia and Liang (2017). No closed-source model or API was used.

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